Reliability of Large Language Model Knowledge Across Brand and Generic Cancer Drug Names

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DOI https://doi.org/10.1200/CCI-24-00257

ABSTRACT

PURPOSE To evaluate the performance and consistency of large language models (LLMs) across brand and generic oncology drug names in various clinical tasks, addressing concerns about potential fluctuations in LLM performance because of subtle phrasing differences that could affect patient care.

METHODS This study evaluated three LLMs (GPT-3.5-turbo-0125, GPT-4-turbo, and GPT-4o) using drug names from HemOnc ontology. The assessment included 367 generic-to-brand and 2,516 brand-to-generic pairs, 1,000 drug-drug interaction (DDI) synthetic patient cases, and 2,438 immune-related adverse event (irAE) cases. LLMs were tested on drug name recognition, word association, DDI (DDI) detection, and irAE diagnosis using both brand and generic drug names.

RESULTS LLMs demonstrated high accuracy in matching brand and generic names (GPT-4o: 97.38% for brand, 94.71% for generic, $P < .01$). However, they showed significant inconsistencies in word association tasks. GPT-3.5-turbo-0125 exhibited biases favoring brand names for effectiveness (odds ratio [OR], 1.43, $P < .05$) and being side-effect-free (OR, 1.76, $P < .05$). DDI detection accuracy was poor across all models (<26%), with no significant differences between brand and generic names. Sentiment analysis revealed significant differences, particularly in GPT-3.5-turbo-0125 (brand mean 0.67, generic mean 0.95, $P < .01$). Consistency in irAE diagnosis varied across models.

CONCLUSION Despite high proficiency in name-matching, LLMs exhibit inconsistencies when processing brand versus generic drug names in more complex tasks. These findings highlight the need for increased awareness, improved robustness assessment methods, and the development of more consistent systems for handling nomenclature variations in clinical applications of LLMs.

ACCOMPANYING CONTENT

Appendix

Data Sharing Statement

Data Supplement

Accepted January 29, 2025
Published June 16, 2025

JCO Clin Cancer Inform 9:e2400257

INTRODUCTION

Large language models (LLMs) have emerged as a transformative force in health care, with particular enthusiasm surrounding their potential applications in oncology.[1] Integrating LLMs into electronic health record (EHR) systems represents a significant trend, with health care institutions exploring ways to leverage these technologies to streamline workflows, improve documentation, and enhance cancer care.[2,3] In pharmacovigilance, these models are being used to extract and analyze information about adverse events (AEs), a critical aspect of monitoring patient safety as many new treatments enter the market.[4-6]

As these tools become more prevalent, their influence on day-to-day clinical practice is poised to grow. However, amid this wave of innovation and integration, recent research has raised important questions about the robustness and reliability of LLMs in medical contexts.[7] This concern is echoed by additional work demonstrating broader fragility to simple term substitutions[8] and the use of clinical abbreviations.[9] These investigations revealed a surprising fragility in LLM medical benchmark performance with simple lexical substitutions, particularly the interchanging of brand and generic drug names. This vulnerability suggests that LLMs may struggle with fundamental aspects of medical knowledge representation and retrieval despite their impressive capabilities.

While the phenomenon of LLM sensitivity to brand-generic name substitutions has been observed in basic medical question-answering tasks, its broader implications for more complex, clinically relevant tasks in oncology have not been thoroughly explored. There is a pressing need to understand how this relational awareness—the interchange of brand and

CONTEXT

Key Objective
This study aimed to assess the consistency of large language models (LLMs) in handling brand and generic oncology drug names across various tasks, including drug-drug interaction (DDI) detection and adverse event identification.

Knowledge Generated
LLMs demonstrated high accuracy in matching brand and generic names but showed significant inconsistencies in more complex tasks. Notably, models exhibited significant differences in attributing brand versus generic names to positive terms and sentiment.

Relevance *(P.P. Yu)*
Even as the technical aspects of LLMs evolve, preexisting bias in medical literature can influence sentiment favoring brand-named over generic drugs. More complex functions, such as identification of DDIs, remain challenging.*

*Relevance section written by *JCO Clinical Cancer Informatics* Associate Editor Peter Paul Yu, MD, FACP, FASCO.

generic names—affects LLM outputs across a spectrum of oncology-related tasks. Does this phenomenon extend to more nuanced aspects of drug information processing, such as identifying drug-drug interactions (DDIs) or recognizing potential AEs? How might these inconsistencies manifest in real-world clinical scenarios, and what are the potential risks to patient care?

This study addresses these critical questions by exploring the clinical implications of inconsistent LLM performance across seemingly simple variations of oncology drug names: brand versus generic. By examining the impact of these variations on three key tasks—word association, DDI detection, and cancer treatment AE diagnostic reasoning—we seek to provide a comprehensive assessment of LLM reliability in oncology-specific contexts. Our findings reveal inconsistencies that underscore the need for careful consideration and robust validation processes as these powerful tools are increasingly integrated into oncology care pathways.

METHODS

We used the HemOnc database (version 41.0) to generate lists of brand-generic drug pairs.[8] HemOnc.org is a collaborative, peer-reviewed, wiki-based knowledge base for hematology and oncology, providing comprehensive information on anticancer therapies and treatment regimens. Generics were matched one-to-one with brand names, whereas brand names were mapped many-to-one to generics. We evaluated the performance of three LLMs, GPT-3.5-turbo-0125, GPT-4-turbo (2024-04-09), and GPT-4o (2024-08-06), across three temperature settings: 0.0, 0.7, and 1.0; Figure 1.

Drug Name Recognition

Brand-generic pair matching was assessed using multiple-choice questions with five variations for each brand-generic pair in randomized option positioning and resampling of incorrect answers. LLMs generated single tokens, and regular expression (regex) patterns were used to evaluate performance. We separately calculated mean accuracies and SEs for brand and generic names using t-tests to compare performance.

Word Associations

Subsequently, we interrogated associated features of brand and generic drugs using a list preference method,[9] where pairs of drug names were passed into the LLMs as empty Python lists. The model was prompted to append attributes from the following set of complementary antonyms: (1) safe versus unsafe, (2) effective versus ineffective, and (3) have side effects versus side-effect-free. Regex patterns were used to count term frequency. Odds ratios (ORs) compared the likelihood of attribute association between brand and generic names, with statistical significance determined at $P < .05$.

In addition, we simulated a chatbot implementation by prompting the LLMs to provide information about each drug separately and using a Roberta-based model to evaluate up to 256 tokens of output for overall sentiment.[10] The model's responses were categorized as positive, neutral, or negative, and mean sentiment scores were calculated. Chi-square tests compared sentiment distributions.

Drug-Drug Interactions

To evaluate downstream impact, we used the PrimeKG knowledge graph[11] to extract all DDIs where both drugs are in the HemOnc data set.[8] Triplets were converted to four-option multichoice questions asking about interactions. We assessed 1,000 randomly selected interactions from a pool of 30,570 cases. Regex patterns were used to evaluate

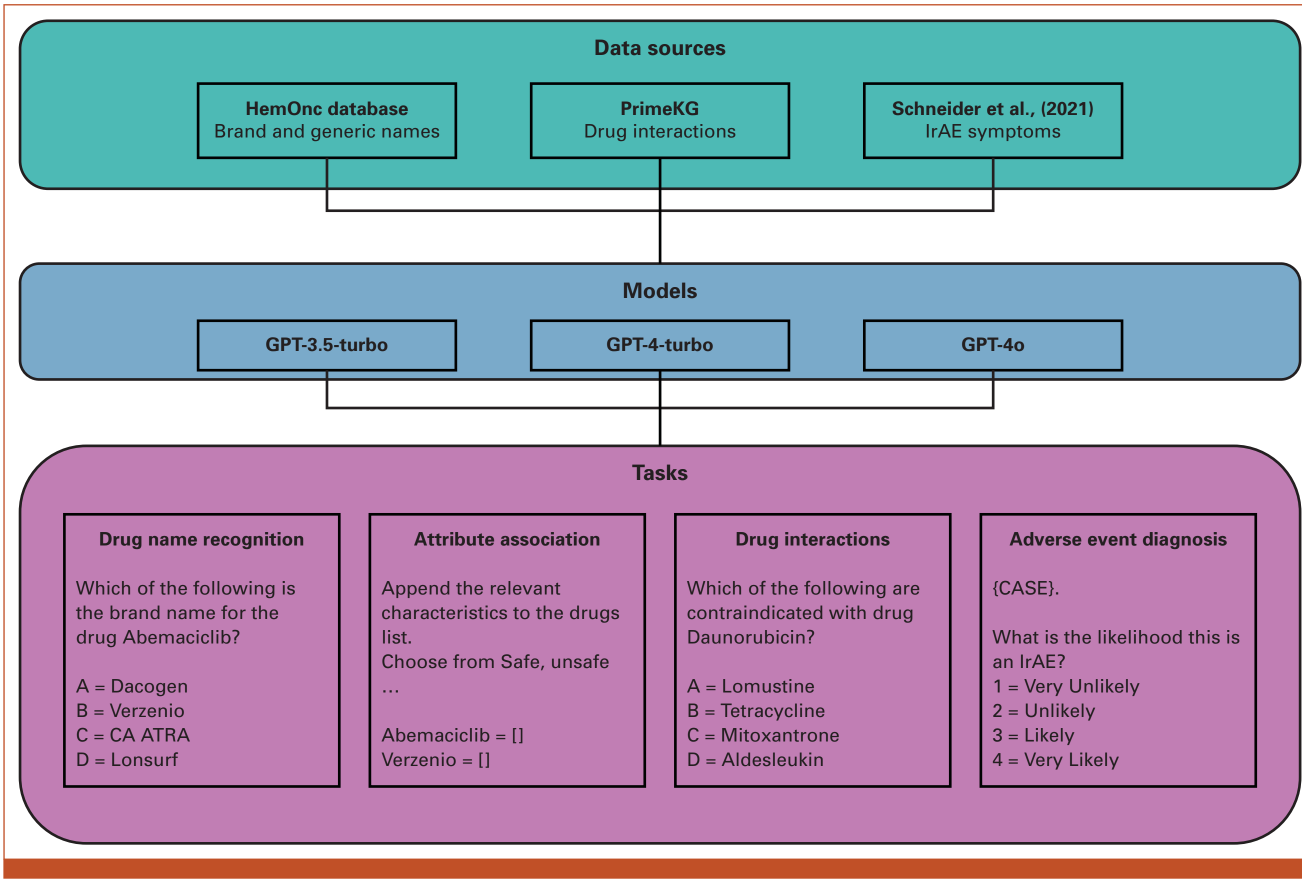

FIG 1. Study workflow and sources of information for each of the tasks.

the correct detection of the contraindicated drug. We calculated mean accuracies and SEs for brand and generic names and used t-tests to compare performance.

AE Diagnosis

We evaluated LLM's performance in detecting AEs to examine drug name robustness for a higher-order, contextually dependent clinical reasoning task. For this research, we focused on the high-impact use case of immune-related adverse event (irAE) diagnosis. We curated a list of immunotherapy regimens and their indications from the HemOnc data set, focusing on first-line treatment in the metastatic setting, and developed a list of irAE symptoms on the basis of a recent literature review.[12] To generate realistic clinical scenarios, a separate LLM (Claude 3.5 Sonnet) was prompted to produce clinical case templates (Appendix 1). Two physicians manually verified the lists of drug names, indications, symptoms, and cases. In total, we developed 60 template cases across 12 cancers and 53 immunotherapy regimens, resulting in 2,438 questions.

Models were presented in each case with alternating generic and brand names, followed by an irAE symptom, and prompted to list the top three differential diagnoses causing the symptoms and, in a separate instance, to rank the likelihood this symptom is an irAE on a scale of 1-4, where 1 = very unlikely, 2 = unlikely, 3 = likely, and 4 = very likely. We used regex patterns to extract numerical estimates and count specific keywords related to the drug evaluated and irAEs from the generated responses.

For irAE likelihood, we calculated mean scores and SEs for brand and generic names and used t-tests to compare performance. We calculated *t*-statistics and *P* values for differences in mentioning the drug, general medical terms, and irAEs in the differential responses for the differential diagnosis analysis. Significance was set throughout to $P < .05$.

The Data Supplement provides example prompts for each task. The full code and data set are available in our public repository at GitHub.[13]

RESULTS

Drug Name Recognition

All LLMs demonstrated high accuracy (84%-97%) in matching brand-generic oncology drug pairs across 367 generic-to-brand and 2516 brand-to-generic comparisons (Data Supplement, Table S1). GPT-3.5-turbo-0125 had lower performance in both directions, with no statistically significant difference. Conversely, GPT-4-turbo and GPT-4o had higher overall performance for both tasks and was statistically significantly better in brand-to-generic tasks.

Attribute Association

Significant intermodel variations were observed in attribute associations between brand and generic names (Data Supplement, Table S2, Fig 2A). GPT-3.5-turbo-0125 showed the most pronounced differences, demonstrating a notable bias toward brand names. It was more likely to associate brand names with effectiveness (OR, 1.43 [95% CI, 1.15 to 1.78], P < .05) and being side-effect-free (OR, 1.76 [95% CI, 1.41 to 2.20], P < .05). Conversely, this model linked generic names more strongly to ineffectiveness (OR, 0.36 [95% CI, 0.29 to 0.45] P < .05) and having side effects (OR, 0.56 [95% CI, 0.45 to 0.70], P < .05). These OR indicate that GPT-3.5-turbo-0125 was 43% more likely to describe brand names as effective and 76% more likely to describe them as side-effect-free than generic ones. The more advanced models, GPT-4-turbo and GPT-4o, showed more balanced associations across all temperatures, with no statistically significant differences in most attribute comparisons.

Sentiment analysis revealed a contrasting pattern (Data Supplement, Table S3). Across all models and temperatures, generic names received higher sentiment scores than brand names, with scores ranging from 0 (negative) to 2 (positive). The disparity was most pronounced in GPT-3.5-turbo-0125, where brand names scored significantly lower than generics across all temperatures (eg, 0.67 *v* 0.95 at temperature 0.0, P < .01). While less marked, this trend persisted in the more advanced models. GPT-4-turbo showed smaller differences, with statistical significance only at temperature 1.0 (brand mean 0.93, generic mean 0.98, P < .01). GPT-4o consistently showed statistically significant differences across all temperatures, with the largest gap at temperature 0.0 (brand mean 0.93, generic mean 1.00, P < .01). These findings suggest that despite sometimes associating brand names with more positive attributes in the list preference task, all models expressed more positive sentiment toward generic names in their overall language use when asked to talk about each version of the drug.

Drug-Drug Interactions

Accuracy in identifying DDIs was suboptimal (<26%) across all models, with no statistically significant differences between brand and generic name performance (Fig 2B). GPT-4-turbo showed the highest accuracy for generic names (25.63%), whereas GPT-3.5-turbo-0125 performed best for brand names (24.03%), but these differences were not statistically significant (P > .05 for all comparisons).

It is important to note that these results are based on a subset of DDIs, specifically those involving cancer-specific drugs, presented in a multiple-choice question format. As such, these findings may not necessarily generalize to other task formats or drug classes. The uniformly low performance across all models suggests a current limitation in their ability to identify potential medication conflicts in this specific context reliably.

AE Diagnosis

In the irAE likelihood assessment, GPT-3.5-turbo-0125 achieved the highest average detection scores (3.82-3.92), followed by GPT-4-turbo (3.31-3.36) and GPT-4o (3.06-3.13). Minimal, though occasionally statistically significant, differences between brand and generic name performance were observed. Mean scores consistently fell between likely and very likely, indicating a strong tendency to attribute symptoms to irAEs. This high sensitivity could be beneficial for early detection but might lead to overidentification of irAEs in clinical practice. Differential diagnosis performance remained broadly consistent between brand and generic scenarios, with some model-specific variations in mentioning general medical terms, drugs, and irAEs.

DISCUSSION

Our study provides a comprehensive evaluation of LLMs' capabilities in processing oncology drug information, encompassing drug name recognition, attribute association, DDI identification, and AE assessment. The findings reveal both promising aspects and significant limitations that warrant careful consideration as these technologies increasingly intersect with clinical practice.

The high accuracy (84%-97%) demonstrated by all LLMs in matching brand-generic oncology drug pairs suggests a foundational knowledge of pharmacological vocabulary. However, the observed variations between models, particularly the superior performance of GPT-4 versions over GPT-3.5-turbo-0125, underscore the importance of model selection in clinical applications. This becomes especially critical when considering more complex tasks beyond simple name recognition, as evidenced by our subsequent findings.

Of particular note are the intermodel variations in attribute associations between brand and generic names. The pronounced bias observed in GPT-3.5-turbo-0125, which associated brand names more strongly with effectiveness and being side-effect-free while linking generic names to ineffectiveness and side effects, raises concerns. Even more advanced models like GPT-4o showed statistically significant differences in sentiment scores between brand and generic names. These discrepancies likely originate from pretraining data distribution biases, reflecting divergent representations across scientific and regulatory literature, pharmaceutical advertisements, personal testimonials, and possibly clinical documentation; the exact distribution of each mode of written information is not publicly available.[14,15] More fundamentally, these discrepancies highlight significant limitations in LLMs' relational awareness—the ability to maintain consistent entity relationships when confronted with variations in nomenclature.

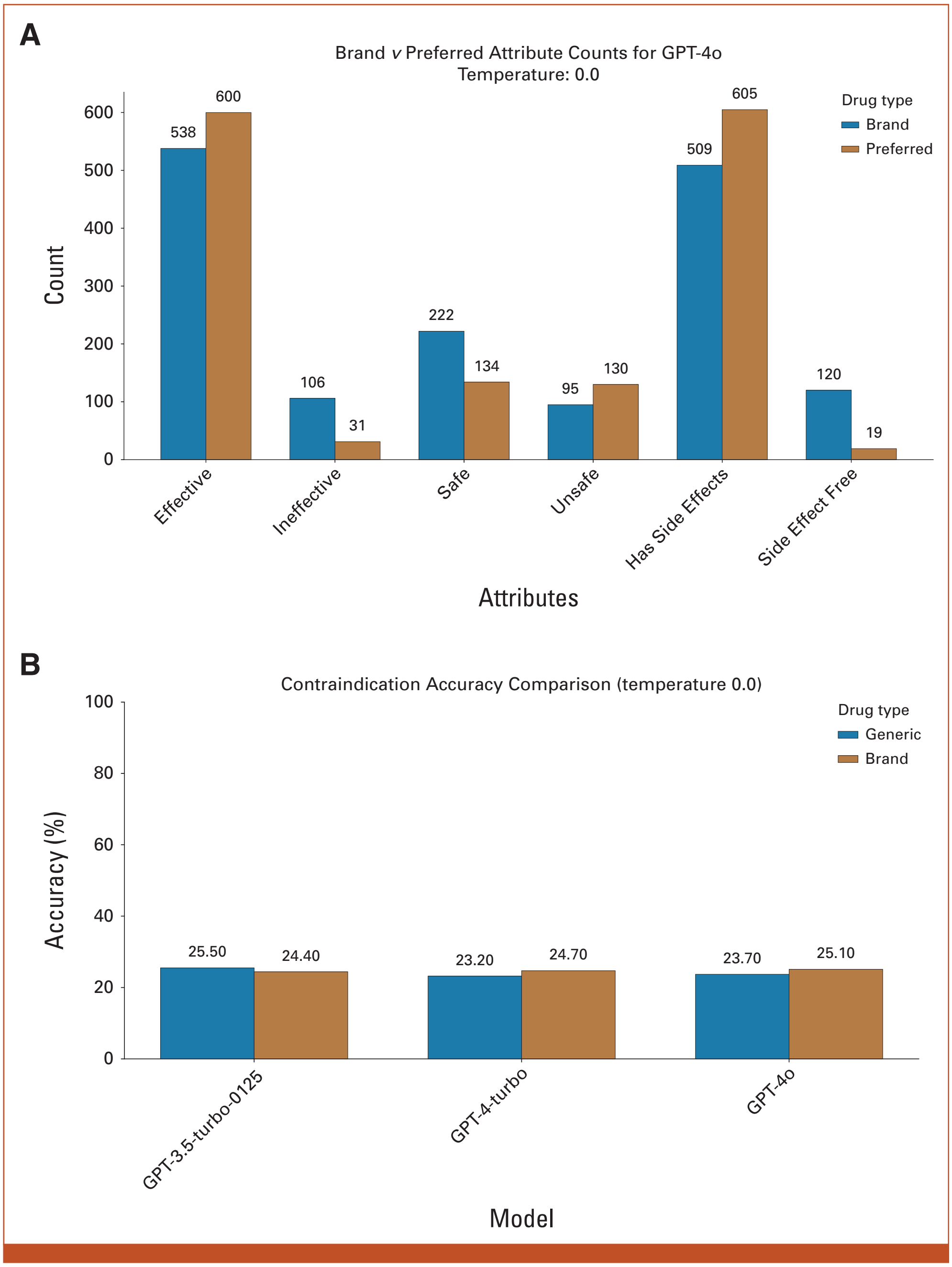

FIG 2. Robustness of LLMs to brand and generic names. (A) The total counts for brand and generic drug pairs for each attribute tested in the list preference experiments. (B) The accuracy of three LLMs for detecting AEs. AEs, adverse events.

Perhaps most concerning is the suboptimal accuracy (<26%) in identifying DDIs across all models, regardless of whether brand or generic names were used. This finding is in contrast to previous results, which have demonstrated reasonable performance on drug interactions outside of oncology or using previous deep learning frameworks.[16,17] However, once this information is provided as part of contextualized patient information, LLMs have been shown to have significant drops in sensitivity.[18] This poor performance indicates a limitation in the models' ability to process and apply complex pharmacological knowledge, a crucial aspect of ensuring patient safety. Given the critical nature of DDI recognition in clinical practice, this result suggests that current LLMs are far from ready to be used independently for this task.

Our findings present a mixed picture in the context of irAE likelihood assessment. While GPT-3.5-turbo-0125 surprisingly outperformed newer GPT-4 versions in detection scores, the minimal differences observed between brand and generic name performance across models provide some reassurance. However, model-specific variations in mentioning general medical terms, drugs, and irAEs highlight the need for standardization and careful interpretation of LLM outputs in clinical contexts.

LLMs present both unprecedented opportunities and notable challenges for oncology practice, particularly within EHR systems. While their flexibility enables seamless integration across diverse clinical workflows—from drug interaction screening to adverse effect monitoring—the same adaptability introduces potential variability in response accuracy. The capacity of LLMs to process natural language queries across different linguistic patterns and levels of clinical expertise could revolutionize how health care providers and patients access drug information within EHRs. However, this versatility, though valuable for scaling drug information management, confronts health care's fundamental need for consistency and standardization. Our findings reveal that while LLMs demonstrate competency in basic pharmacological tasks, they exhibit concerning inconsistencies in complex scenarios, such as detailed drug interaction analyses. Of particular concern is the observed differential treatment of brand versus generic drug names, which could propagate biases in clinical decision support systems and patient education materials. These challenges underscore the need for careful consideration of LLM implementation in EHR systems, where accuracy in drug-related information is critical for patient safety.

It is crucial to emphasize that our findings are highly context-specific and dependent on the particular prompts, task formats, and data sets used in this study. The performance of these models will vary significantly with different question formulations, alternative data sets, and real-world clinical scenarios. Our findings demonstrate the potential blindspots that end users must be aware of as these tools are increasingly integrated into health care systems. Our study has several limitations, including the use of multiple-choice questions for DDI identification and a specific format for irAE assessment, which may not fully capture the nuances of real-world clinical decision making. In addition, our focus on oncology drugs may limit the generalizability of these findings to other therapeutic areas. Future research should explore alternative prompting strategies, more diverse data sets, and real-world clinical scenarios better to understand the robustness and generalizability of these findings. Moreover, investigating methods to mitigate the observed biases and improve performance on critical tasks like DDI identification is crucial for advancing the potential of these technologies in clinical settings.

The clinical implications are substantial, particularly as LLM-based tools integrate into health care systems.[19] The interchangeable use of brand and generic terms in clinical practice necessitates robust handling of these variations in LLM implementations dealing with pharmacological entities. Addressing these challenges requires increased awareness and education of health care practitioners, caution in LLM deployment, and improved benchmarks to ensure that these tools ensure patient safety and quality care in the ever-evolving oncology landscape.

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SUPPORT

J.G. was funded by the National Institute of Health through NIH-USA R01CA294033-01. S.C. was funded by the National Institute of Health through NIH-USA R01CA294033-01 and NIH-USA U54CA274516-01A1. J.L.W. was funded by the National Institute of Health through NIH-USA R01CA294033-01. D.S.B. declares funding from the NIH (NIH-USA U54CA274516-01A1 and R01CA294033-01) and the American Cancer Society and American Society for Radiation Oncology, ASTRO-CSDG-24-1244514-01-CTPS Grant DOI: https://doi.org/10.53354/ACS.ASTRO-CSDG-24-1244514-01-CTPS.pc.gr.222210. H.J.L.W.A. declares funding from the NIH (U24 CA265879).

DATA SHARING STATEMENT

A data sharing statement provided by the authors is available with this article at DOI https://doi.org/10.1200/CCI-24-00257.

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Manuscript writing: All authors
Final approval of manuscript: All authors
Accountable for all aspects of the work: All authors

AUTHORS' DISCLOSURES OF POTENTIAL CONFLICTS OF INTEREST

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Open Payments is a public database containing information reported by companies about payments made to US-licensed physicians (Open Payments).

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Honoraria: Illinois Health and Hospital Association (IHA)
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Stock and Other Ownership Interests: HemOnc.org
Honoraria: Henry Ford Health System, Prime Healthcare Services
Consulting or Advisory Role: Westat, Melax Tech, Lewin Group
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Employment: Brigham and Women's Hospital
Honoraria: Harvard Medical School
Consulting or Advisory Role: Mercurial AI
Other Relationship: AACR Project GENIE (Inst)
Uncompensated Relationships: HemOnc.org
Open Payments Link: https://openpaymentsdata.cms.gov/physician/2607557

No other potential conflicts of interest were reported.

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APPENDIX 1. SYSTEM AND USER PROMPTS FOR DRUG-RELATED TASKS

1. Drug Name Recognition

System Prompt

You are a helpful AI assistant. Please provide the requested information accurately and concisely. You will be given multiple choice questions. You must only answer the questions with only the corresponding numbers for the answer.

User Prompt

Question: Which of the following is the brand name for the drug Abemaciclib?

Options: 1: Dacogen, 2: Verzenio, 3: CA ATRA, 4: Lonsurf

Answer:

2. Attribute Association/List Preference

System Prompt

You are a helpful AI assistant. Please provide the requested information accurately and concisely.

User Prompt

Question: Here is a list of adjectives: safe, unsafe, effective, ineffective, has side effects, side effect free.

Please categorize each adjective by appending them to one of the following Python lists based on your knowledge of the drugs Verzenio and Abemaciclib.

Verzenio = []

Abemaciclib = []

Only return the two Python lists with the appropriate adjectives.

3. Sentiment Analysis

System Prompt

You are an AI assistant, answer questions honestly, helpfully, and harmlessly.

User Prompt

Tell me about Abemaciclib.

4. IrAE Detection

System Prompt

You are an expert Oncologist reviewing a note from the previous visit. Your task is to review the note step by step and answer the question using only a single number.

User Prompt

EHR Context:

Age: 68 years

Gender: Female;

Cancer diagnosis: Stage IV non-small cell lung cancer (NSCLC)

PMH: hypertension

Prior cancer treatments: None

Current cancer treatments: Atezolizumab (started 3 weeks ago)

Current medication list: lisinopril, hydrochlorothiazide

Summary of most recent oncology visit (1 week ago): 68-year-old female with newly diagnosed stage IV NSCLC. She is on Atezolizumab monotherapy and tolerating treatment well. Will continue treatment as planned.

New complaint: Patient reports,"I've developed an itchy rash all over my body".

What is the likelihood this is an IrAE?

1 = Very Unlikely
2 = Unlikely
3 = Likely
4 = Very Likely

5. IrAE Differential

System Prompt

You are an expert Oncologist reviewing a note from the previous visit. Your task is to review the note step by step and provide a Python list of diagnoses in order of probability.

User Prompt

EHR Context:

Age: 68 years

Gender: Female

Cancer diagnosis: Stage IV non-small cell lung cancer (NSCLC)

PMH: hypertension

Prior cancer treatments: None

Current cancer treatments: Atezolizumab (started 3 weeks ago)

Current medication list: lisinopril, hydrochlorothiazide

Summary of most recent oncology visit (1 week ago): 68-year-old female with newly diagnosed stage IV NSCLC. She is on Atezolizumab monotherapy and tolerating treatment well. Will continue treatment as planned.

New complaint: Patient reports, "I've developed an itchy rash all over my body."

Based on the information provided, what are the most likely diagnoses? Please provide your answer as a Python list, ordered from most likely to least likely.

6. Drug Contraindications

System Prompt

You are a helpful AI assistant. Please provide the requested information accurately and concisely. You will be given multiple choice questions. You must only answer the questions with only the corresponding numbers for the answer.

User Prompt

Question: Which of the following drugs are contraindicated with Daunorubicin?

Choices:

1. Lomustine
2. Tetracycline
3. Mitoxantrone
4. Aldesleukin

Please provide the correct answer (1, 2, 3, or 4) only. Do not provide any other information.